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Submitted to Dr. Mohsen Rashwan

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Table 1: Team Members

1 Executive Summary

1.1 Overview of the Startup Idea

Our startup, *SurgeAI*, is set to transform recruitment and career preparation in the Middle East and North Africa (MENA) region with an AI-powered interviewer built on Large Language Models (LLMs) tailored for Arabic users. *SurgeAI* serves two core markets: for companies, it automates candidate screenings to streamline hiring; for individuals, it offers mock interview practice to boost job readiness. By integrating advanced natural language processing with deep cultural customization, *SurgeAI* addresses the unique linguistic and professional needs of Arabic-speaking users, a market overlooked by global platforms.

The MENA region faces significant challenges in recruitment and career development. Companies in key markets like Saudi Arabia, the United Arab Emirates (UAE), and Egypt struggle with high applicant volumes and time-intensive interview processes. Meanwhile, job seekers, from fresh graduates to mid-career professionals, lack accessible tools to practice interviews in Arabic that align with regional workplace norms. Driven by a growing workforce—supported by initiatives like Saudi Arabia’s Vision 2030—and increasing AI adoption, *SurgeAI* is positioned to meet a critical market need.

1.2 Problem Statement

Recruitment in the MENA region is often inefficient and lacks localization. Companies face difficulties screening large candidate pools quickly while ensuring culturally appropriate evaluations. Global AI interview platforms, such as HireVue or InterviewBuddy, provide limited or no Arabic language support and fail to address regional dialects and cultural nuances, rendering them ineffective for MENA markets. Job seekers lack Arabic-specific mock interview tools to prepare for interviews reflecting local expectations, such as formal communication styles or sector-specific questions (e.g., technology, finance, energy). Regional job platforms like Bayt.com and Wuzzuf focus on job listings and resumes, not AI-driven interview solutions. *SurgeAI* bridges this gap with a tailored, scalable product.

1.3 Proposed Solution

SurgeAI delivers a versatile AI-powered interviewer designed for corporate hiring teams and individual job seekers. For companies, it automates initial interviews with customizable question sets, real-time response evaluations, and seamless HR system integrations, reducing hiring costs and time. For individuals, it provides mock interviews with personalized feedback on content, tone, and cultural appropriateness, tailored to specific roles or industries. The platform supports Arabic (with adaptability for Gulf, Egyptian, and Levantine dialects) and English, ensuring broad accessibility across MENA markets.

Key features include:

- **Corporate Use:** Automated screening, role-specific question generation, candidate scoring, and HR platform integrations.
- **Individual Use:** Mock interviews, detailed feedback, video recording options, and industry-specific scenarios.
- **Cultural Relevance:** Questions and feedback aligned with Arabic workplace norms and regional dialects.

The platform will be accessible via web and mobile apps, featuring a user-friendly interface with voice and text interaction options to maximize engagement.

1.4 Market Analysis: Key Findings

Market research underscores strong demand for an AI-powered interviewer in the MENA region. The recruitment market is expanding due to economic diversification and youth employment initiatives, with Saudi Arabia, UAE, and Egypt as key hubs. Surveys indicate that 70% of HR professionals seek tools to automate initial screenings, while 65% of job seekers (aged 18–35) express interest in Arabic-language interview practice platforms. Competitor analysis reveals a lack of Arabic-specific AI interview tools, with global platforms failing to address linguistic and cultural needs and regional job platforms focusing on non-AI services.

The target audience includes:

- **Companies:** HR departments in SMEs and enterprises, particularly in technology, finance, and energy sectors.
- **Individuals:** Job seekers, including students, graduates, and professionals, pursuing career advancement.
- **Educational Institutions:** Universities and career centers offering job preparation services.

Initial focus will target Saudi Arabia, UAE, and Egypt, with plans to expand to Qatar, Jordan, and Arabic-speaking diaspora communities.

1.5 Unique Value Proposition

SurgeAI is the first AI-powered interviewer designed specifically for Arabic users, blending linguistic precision with cultural relevance. Unlike global competitors, it offers:

- **Arabic Customization:** Support for regional dialects and workplace norms, ensuring natural and relevant interactions.
- **Dual-Purpose Design:** A single platform serving both corporate hiring and individual career preparation.
- **Scalability:** Cloud-based infrastructure and integrations for seamless adoption by businesses and users.

By addressing the underserved needs of the MENA recruitment and career preparation market, *SurgeAI* is poised to capture significant market share.

1.6 Vision and Next Steps

Our vision is to empower MENA's workforce and businesses by making recruitment and career preparation more efficient, accessible, and culturally aligned. *SurgeAI* will launch a minimum viable product (MVP) within six months, focusing on text-based mock interviews and basic corporate screening features. Beta testing with local universities and SMEs will follow, with a full launch planned within 18 months. Strategic partnerships with job platforms (e.g., Bayt, Wuzzuf) and educational institutions will drive adoption, while a freemium model for individuals and subscription plans for businesses will ensure sustainable revenue.

SurgeAI is well-positioned to lead the AI-driven recruitment and career preparation market in MENA, delivering transformative value to companies, job seekers, and society.

2 Market Analysis

2.1 Market Need & Opportunity

The Middle East and North Africa (MENA) region offers a compelling opportunity for *SurgeAI*, an AI-powered interviewer leveraging Large Language Models (LLMs) tailored for Arabic users. Recruitment processes in MENA are often inefficient, with companies in Saudi Arabia, the United Arab Emirates (UAE), and Egypt struggling to manage high applicant volumes. A 2023 survey by Panorama RH in Morocco revealed that 70% of HR professionals seek automation to streamline initial screenings, a sentiment echoed across MENA’s corporate landscape [16]. For job seekers, particularly young professionals aged 18–35, the absence of Arabic-language interview preparation tools hinders their ability to excel in interviews aligned with regional workplace norms, such as formal communication, hierarchical respect, and sector-specific expertise (e.g., technology, finance, energy).

The need for an Arabic LLM-based solution is underscored by MENA’s linguistic diversity. Arabic, spoken by over 400 million people, features dialects like Gulf, Egyptian, and Levantine, which vary significantly in vocabulary and tone [9]. Global AI interview platforms, such as HireVue or InterviewBuddy, lack Arabic support and cultural customization, rendering them ineffective for MENA markets. Regional job platforms like Bayt.com and Wuzzuf focus on job listings and resume-building, not AI-driven interview solutions. *SurgeAI* addresses this gap with a dual-purpose platform: automated candidate screening for companies and mock interview practice for individuals, both tailored to Arabic dialects and cultural nuances.

Economic and demographic trends amplify this opportunity. Saudi Arabia’s Vision 2030, UAE’s AI Strategy 2031, and Egypt’s Digital Transformation Strategy prioritize workforce development and technology adoption, driving demand for efficient hiring tools [17]. The MENA recruitment market is projected to grow from USD 58.25 billion in 2022 to USD 132.18 billion by 2030, with a CAGR of 10.8% [5]. High youth unemployment rates (e.g., 32.6% in Egypt, 26.7% in Saudi Arabia) highlight the need for career preparation tools to enhance employability [17]. Furthermore, the rise of remote work and digital hiring platforms post-COVID has accelerated the adoption of AI in recruitment, with 65% of MENA companies planning to invest in HR tech by 2026 [1]. *SurgeAI* is well-positioned to capitalize on these trends by delivering a scalable, culturally relevant solution.

2.2 Target Audience

SurgeAI targets three primary segments, each defined by demographics, behavioral patterns, and adoption potential:

- **Corporate HR Departments (B2B):**
 - *Demographics:* HR managers and recruiters, aged 25–50, in SMEs and large enterprises across Saudi Arabia, UAE, and Egypt. Key sectors include technol-

ogy (e.g., software development), finance (e.g., banking), energy (e.g., oil and gas), and healthcare.

- *Behavioral Patterns*: Prioritize tools that reduce hiring time and costs, seek seamless integration with HR platforms (e.g., SAP SuccessFactors, Workable), and value data-driven insights for candidate evaluations. A 2023 IDC report indicates that 25% of MENA’s AI investment targets financial services, signaling strong adoption potential in this sector [17].
- *Adoption Rates*: High, with 70% of HR professionals expressing interest in AI automation [16]. SMEs, constrained by resources, are likely to adopt quickly, while enterprises may require customized integrations and proof of ROI.

- **Individual Job Seekers (B2C):**

- *Demographics*: Students, fresh graduates, and mid-career professionals, aged 18–35, in urban centers like Riyadh, Dubai, and Cairo. Gender split is approximately 60% male, 40% female, reflecting MENA workforce trends.
- *Behavioral Patterns*: Actively engage with mobile apps and online platforms (e.g., LinkedIn, Bayt) for career development, seek affordable interview practice tools, and value personalized feedback on communication, tone, and cultural appropriateness. Social media platforms like Instagram and TikTok drive awareness among younger users.
- *Adoption Rates*: Moderate to high, with 65% of job seekers interested in Arabic-language career tools [16]. A freemium model with basic mock interviews can drive initial adoption, with premium features (e.g., video recording, advanced analytics) encouraging upgrades.

- **Educational Institutions (B2B2C):**

- *Demographics*: Career centers and universities serving students aged 18–25, including King Saud University (Saudi Arabia), UAE University, and Cairo University.
- *Behavioral Patterns*: Seek scalable, cost-effective tools to prepare students for competitive job markets, value partnerships with tech providers, and prioritize user-friendly interfaces. Demand is driven by government initiatives like Egypt’s 2030 Education Strategy, which emphasizes employability.
- *Adoption Rates*: Moderate, as universities adopt technology slower than corporates but are motivated by youth employment goals. Annual contracts with tiered pricing can accelerate adoption.

Adoption barriers include cost sensitivity among individual users and regulatory compliance concerns for companies. *SurgeAI* will mitigate these through tiered pricing models

(e.g., freemium for individuals, subscriptions for businesses) and robust data governance aligned with MENA regulations.

2.3 Competitive Analysis

SurgeAI operates in a niche with few direct competitors but faces competition from global AI platforms, regional job boards, and traditional recruitment methods. Below is an expanded analysis of competitors, followed by a SWOT table and a competitor comparison table.

Competitors and Substitutes:

- **Global AI Interview Platforms:**

- ***HireVue***: Offers AI-driven video interviews with candidate scoring but lacks Arabic support and MENA-specific customization. Targets large enterprises globally [11].
- ***Qureos***: A MENA-focused AI recruitment platform with video assessments, candidate sourcing, and screening, tailored for GCC recruiters. Supports Arabic, making it a direct competitor [18].
- ***CVViZ***: A global AI tool for candidate engagement, scheduling, and ATS integration, competing with *SurgeAI*'s B2B features. Not MENA-specific but relevant for enterprises [7].
- ***InterviewBuddy***: Provides mock interview practice for individuals, primarily in English, competing with *SurgeAI*'s B2C feature but lacking Arabic support [12].
- ***Avature***: A global recruitment platform with AI-driven candidate management, less MENA-focused but a competitor for corporate clients [3].

- **Regional Job Platforms:**

- ***Wuzzuf***: Egypt-based platform for job listings and applicant tracking, a substitute for corporate hiring but lacking AI interview features [20].
- ***Bayt.com***: A leading MENA job board with resume-building tools, similar to Wuzzuf, with no AI interview capabilities [4].
- ***Akhtaboot***: Jordan-based job platform operating across MENA, offering recruitment services but no AI-driven interview solutions [2].

- **Manual Recruitment Processes**: Traditional methods (e.g., in-person interviews, manual resume reviews) dominate SMEs in MENA but are time-intensive, costly, and prone to bias.

Competitor Comparison Table:

Feature	SurgeAI	Qureos	HireVue	Wuzzuf
Dialect Support	Yes	Yes	No	Limited
AI-Interview-Automation	Yes	Yes	Yes	No
Mock Interview Practice	Yes	Partial	No	No
Cultural-Customization	High	Moderate	Low	Low
HR Platform Integration	Yes	Yes	Yes	Partial
Target Market	MENA	GCC	Global	MENA

Table 2: Competitor Feature Comparison

SWOT Analysis for SurgeAI:

Strengths	Weaknesses
• First-mover advantage in Arabic-specific AI interviewer market. • Dual-purpose platform for B2B and B2C users, enhancing versatility. • Cultural and dialectal customization (e.g., Gulf, Egyptian, Levantine). • Scalable cloud-based infrastructure with HR integrations.	• Limited brand recognition as a new entrant. • High initial development costs for Arabic LLM training. • Dependence on quality and availability of Arabic datasets. • Potential user skepticism about AI fairness and accuracy.
Opportunities	Threats
• Growing AI adoption in MENA (e.g., UAE AI market CAGR of 43.9% [10]). • Partnerships with universities and job platforms (e.g., Bayt, Wuzzuf). • Expansion to Arabic-speaking diaspora in Europe and North America. • Support from government initiatives (e.g., Saudi Vision 2030).	• Emerging Arabic LLMs (e.g., Jais, Falcon) may enable competitors. • Regulatory ambiguity around AI and data privacy in MENA [1]. • Global players adapting to include Arabic support. • Talent shortage for AI development in MENA [1].

Table 3: SWOT Analysis of SurgeAI

SurgeAI's Arabic customization and dual functionality provide a competitive edge, but it must address data limitations, regulatory compliance, and potential market entry by global players to sustain its lead.

2.4 Market Trends

Several trends in AI adoption and digital transformation in MENA bolster *SurgeAI*'s market potential:

- **Rising AI Adoption:** The MENA AI market is expanding rapidly, with the UAE's AI market valued at USD 3.47 billion in 2023 and projected to grow at a CAGR of 43.9% through 2030 [10]. Generative AI could contribute USD 23.5 billion annually

to MENA’s economy by 2030 [1]. Saudi Arabia’s National Transformation Programme 2020 and UAE’s AI Strategy 2031 prioritize AI-driven innovation, creating a favorable environment for *SurgeAI* [17]. For example, Saudi Arabia aims to train 20,000 AI specialists by 2030, signaling strong governmental support [1].

- **Arabic LLM Development:** Advancements in Arabic LLMs, such as UAE’s Falcon and G42’s Jais, reflect growing investment in language-specific AI. However, these models target general applications (e.g., translation, chatbots), not recruitment, leaving a niche for *SurgeAI* [8]. Challenges include limited Arabic training data (only 0.6% of web content is in Arabic) and dialectal variations, which *SurgeAI* will address through targeted fine-tuning [8].
- **Digital Transformation in Recruitment:** The global AI recruitment market is projected to grow from USD 661.56 million in 2023 to USD 1,119.80 million by 2030, with MENA following suit [13]. Sectors like finance (25% of MENA AI investment), healthcare, and energy are early adopters, aligning with *SurgeAI*’s target industries [17]. The shift to remote hiring post-COVID has further accelerated demand for AI-driven tools, with 60% of MENA HR leaders planning to adopt AI by 2025 [1].
- **Youth Workforce and Education:** MENA’s young population (over 50% under 30) drives demand for career preparation tools. Government initiatives, such as Egypt’s 2030 Education Strategy and UAE’s National Employment Program, emphasize employability, supporting *SurgeAI*’s B2C and B2B2C segments [17]. For instance, Egypt aims to train 1 million youth in digital skills by 2030, creating a ready user base for mock interview tools.

These trends highlight the strategic timing of *SurgeAI*’s market entry, leveraging AI advancements and regional workforce priorities.

2.5 Regulatory & Ethical Considerations

Deploying an AI-powered interviewer in MENA requires navigating a complex landscape of legal, privacy, and ethical challenges, particularly for Arabic language data:

- **Regulatory Landscape:**
 - *Data Privacy:* MENA countries are strengthening data protection frameworks. UAE’s Personal Data Protection Law (PDPL) and Saudi Arabia’s PDPL mandate strict consent, encryption, and anonymization for handling candidate data [1]. Egypt’s Data Protection Law (2020) imposes similar requirements, with fines up to EGP 5 million for violations. *SurgeAI* will implement end-to-end encryption, secure cloud storage (e.g., AWS), and user consent protocols to ensure compliance.
 - *AI Regulation:* MENA lacks a unified AI governance framework, creating ambiguity. The UAE’s AI and Blockchain Council and Digital Dubai’s Ethical AI

Toolkit emphasize transparency, fairness, and accountability, while Saudi Arabia’s AI Ethics Principles (2023) prioritize human oversight [10, 6]. *SurgeAI* will align with these guidelines by adopting explainable AI and regular audits.

- **Ethical Considerations:**

- *Bias Mitigation:* Arabic LLMs risk perpetuating biases if trained on unrepresentative datasets (e.g., gender or dialect imbalances). *SurgeAI* will source diverse datasets from academic partners and crowdsourcing, conduct bias audits, and use fairness-aware algorithms to ensure equitable candidate evaluations [8].
- *Transparency:* LLMs’ opaque decision-making can erode trust, especially in recruitment. *SurgeAI* will implement explainable AI techniques (e.g., feature importance scores) to clarify how responses are scored, aligning with MENA’s emerging AI ethics standards [6].
- *Cultural Sensitivity:* Arabic text carries nuanced cultural meanings that LLMs may misinterpret (e.g., formal vs. colloquial tones). *SurgeAI* will fine-tune its model with region-specific data, validated by linguists, to ensure culturally appropriate interactions [8].
- *Data Security:* Handling sensitive candidate data (e.g., interview recordings) requires robust cybersecurity. *SurgeAI* will comply with international standards (e.g., GDPR, ISO 27001) and conduct regular penetration testing to safeguard user information.

- **Implementation Strategy:** *SurgeAI* will collaborate with local regulatory bodies (e.g., UAE’s Telecommunications and Digital Government Regulatory Authority), partner with universities for ethical data sourcing, and establish an ethics advisory board. Transparent user agreements and opt-in consent will build trust, while compliance with GDPR will facilitate expansion to Arabic-speaking diaspora markets.

Proactively addressing these considerations will ensure *SurgeAI*’s credibility, regulatory compliance, and long-term success in MENA.

3 Product/Service Description

3.1 Concept & Features

SurgeAI is an AI-powered interviewer platform aimed at transforming recruitment and career preparation in the Middle East and North Africa (MENA) region, with a focus on Arabic-speaking users. Built on Large Language Models (LLMs), *SurgeAI* provides a dual-purpose solution: for companies, it automates candidate screening to streamline hiring processes; for individuals, it offers mock interview practice to enhance job readiness. The platform addresses a critical gap in the MENA market by supporting Arabic language and cultural nuances, which global competitors like HireVue and InterviewBuddy fail to address [11, 12]. While the Minimum Viable Product (MVP) demo focuses on text-based interactions, speech-to-text (STT), text-to-speech (TTS), and the Egyptian dialect, the full vision includes support for multiple Arabic dialects (Gulf, Egyptian, Levantine) and advanced features like voice and video interactions. The

MVP demo, as implemented, uses a multi-agent AI workflow to process job details and generate tailored interview questions in professional Egyptian Arabic, with an AI-led interview conducted via Google Meet. The workflow, illustrated in Figure 1, includes five core agents, with an additional validation agent to ensure quality. The process begins with *The Interviewer*, which provides user inputs (job title, requirements) to the *Agent (Interview Script Writer)*. The workflow then proceeds as follows:

- **Agent A (Input Processor):** Analyzes job details provided through the *Agent (Interview Script Writer)* to identify the technical level (entry, mid, senior), key skills, problem-solving relevance, and domain knowledge, outputting a structured JSON.
- **Agent B (Search Query Generator):** Takes the processed input from Agent A and generates up to 10 search queries for interview questions relevant to the job.
- **Agent C (Research Agent):** Uses the search queries from Agent B to perform web searches via a search engine tool, retrieving the best results based on a relevance score threshold (e.g., 0.7) and prioritizing MENA-specific content.
- **Agent D (Web Scraper):** Takes URLs from Agent C, scrapes them using a web scraping tool, and extracts skills, technologies, and example interview questions, ensuring at least one data type is non-empty.
- **Agent E (Question Generator):** Uses the scraped content from Agent D to craft 5–10 interview questions in Egyptian Arabic, categorized by type (technical, problem-solving, scenario-based) and difficulty (easy, medium, hard).
- **Agent G (Validator):** Validates the generated questions from Agent E, providing feedback to ensure relevance and cultural appropriateness before finalizing the output.

The finalized questions are returned to the *Agent (Interview Script Writer)*, which compiles the interview script and delivers it back to *The Interviewer*. For the AI-led interview, the script is used by an AI agent equipped with STT and TTS capabilities, joining the

candidate in a Google Meet session to conduct the interview in Egyptian Arabic, with responses recorded for manual evaluation. Key features of the MVP demo include:

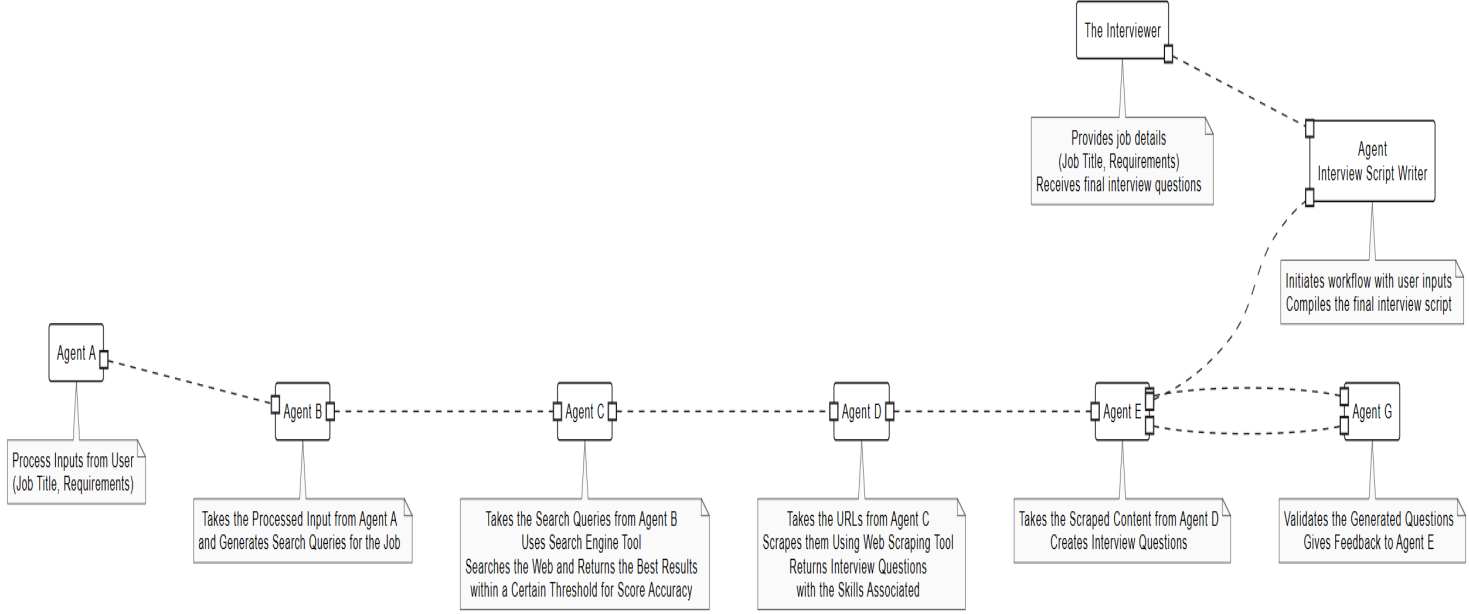


Figure 1: Multi-Agent Workflow for SurgeAI MVP Demo

- **Automated Candidate Screening (B2B):** Generates role-specific questions and provides manual scoring for candidate responses, integrating with HR platforms like Workable [20].
- **Mock Interview Practice (B2C):** Offers text-based and voice-based mock interviews in Egyptian Arabic via Google Meet, where an AI agent uses STT to process candidate responses and TTS to deliver questions, with feedback on content and tone (e.g., for roles like R&D AI ML Developer Intern). Responses are recorded for manual evaluation.
- **Email Notifications:** Sends users an email with generated questions and a Google Meet link for the AI-led mock interview, using Gmail’s SMTP server.

The full vision of *SurgeAI* expands on the MVP to serve a broader MENA audience with the following features:

- **Multi-Dialect Support:** Extends language support to Gulf and Levantine dialects, ensuring accessibility across Saudi Arabia, UAE, and Jordan, in addition to Egypt.
- **Automated Scoring:** Upgrades from manual to automated response evaluation using LLMs, providing detailed analytics on candidate performance for corporate users, including voice-based responses from Google Meet interviews.
- **Voice and Video Interactions:** Enhances voice-based mock interviews with video recording options in Google Meet, simulating real interview scenarios and supporting diverse user preferences.
- **Cultural Customization:** Aligns questions and feedback with regional workplace norms, such as formal communication styles in Gulf countries or sector-specific expectations (e.g., energy sector in Saudi Arabia).

- **Industry-Specific Scenarios:** Generates tailored scenarios for key MENA industries like technology, finance, and energy, enhancing relevance for both corporate and individual users.

This phased approach ensures the MVP validates core functionality in Egypt, while the full vision positions *SurgeAI* as a market leader in AI-driven recruitment across MENA.

3.2 User Experience & Design

SurgeAI delivers a user-friendly experience tailored to the needs of corporate HR professionals, individual job seekers, and educational institutions in the MENA region. The MVP demo is a web-based application with a simple interface zbud using React and Next.js, ensuring accessibility on both desktop and mobile devices. The full vision includes mobile apps and voice-enabled features to enhance engagement across diverse dialects. In the MVP demo, the user journey for job seekers begins with an input form

(implemented using ipywidgets) where users enter job details (e.g., position: R&D AI ML Developer Intern, requirements: Python, TensorFlow experience), company name, score threshold for search relevance (default 0.7), and email. After submission, the platform processes the inputs through its multi-agent workflow, generating 5–10 interview questions in Egyptian Arabic. Users receive an email with the questions and a Google Meet link for a voice-based mock interview, where an AI agent uses STT to process responses and TTS to deliver questions in Egyptian Arabic. Responses are recorded during the Google Meet session for manual evaluation, with text-based feedback provided on content and tone. The interface supports right-to-left Arabic text and prioritizes simplicity for users in urban centers like Cairo. For corporate HR users, the MVP offers a

dashboard to input job requirements and review candidate responses (text and recorded voice) with manual scoring. Recorded responses from Google Meet interviews are accessible for evaluation, and results can be exported to platforms like Workable, ensuring seamless integration. Questions are presented in Egyptian Arabic, aligning with local professional norms. The full vision enhances the user experience by supporting Gulf and

Levantine dialects, allowing users in Saudi Arabia or Jordan to interact in their native dialects. Voice interactions via Google Meet are upgraded with automated scoring, providing real-time feedback on tone, pronunciation, and content. Video recording options in Google Meet allow job seekers to review their performance, addressing the needs of MENA’s tech-savvy youth (over 50% under 30 [17]). The interface will remain intuitive, with multilingual support and culturally relevant design elements (e.g., formal tone for Gulf users). User journey maps will be updated to include voice and video flows, ensuring a seamless experience across all features.

3.3 Technical Roadmap

SurgeAI's technical architecture is designed for scalability and cultural relevance, evolving from the MVP demo to the full vision over an 18-month timeline. The roadmap outlines the architecture, data requirements, model training strategies, and integration plans, focusing on initial deployment in Egypt and expansion across MENA.

Architecture Overview: The MVP demo operates on Amazon Web Services (AWS) with a multi-agent system built using Python and CrewAI, as shown in Figure 1. The frontend uses React and Next.js, communicating with the backend via REST APIs. Outputs, including recorded Google Meet responses, are stored in AWS S3 as JSON and audio files. STT (e.g., Whisper) and TTS (e.g., AWS Polly) modules are integrated to enable voice-based interviews in Google Meet, with the AI agent joining via Google Meet APIs.

The full vision enhances this architecture by adding video handling capabilities (e.g., AWS Media Insights for Google Meet recordings) and a dialect-adaptive LLM layer to route inputs to the appropriate model (Egyptian, Gulf, Levantine), ensuring seamless multi-dialect support. Scalability is maintained through AWS Lambda for serverless processing and AWS RDS for user data management, with audio and video recordings securely stored.

Data Requirements: The MVP demo requires 50,000 Arabic text samples and 10,000 audio samples focused on the Egyptian dialect, sourced from university partnerships (e.g., Cairo University, \$3,000) and web scraping via crawl4ai (e.g., from Wuzzuf [20]). Audio samples support STT and TTS training for Google Meet interviews. The full vision demands an additional 100,000 text samples and 20,000 audio samples per dialect (Gulf, Levantine), totaling 250,000 text and 50,000 audio samples, to ensure robust language and voice support. Data will include job postings, interview guides, tech forums, and audio recordings, validated for cultural relevance. Compliance with Egypt's Data Protection Law (2020) and Saudi Arabia's PDPL is ensured through encryption and anonymization [1].

Model Training and Adaptation: The MVP demo fine-tunes a pre-trained Arabic LLM (e.g., AraBERT) for Egyptian Arabic and a voice model (e.g., Whisper for STT, AWS Polly for TTS) using PyTorch on AWS EC2 (cost: \$20,000), targeting 85% accuracy for dialect-specific text and voice responses. Two LLMs are used: GPT-4o-mini for Agents A–D and GPT-4o for Agent E to balance cost and quality. The full vision involves training separate models for Gulf and Levantine dialects, leveraging transfer learning to minimize data needs. Voice models will be fine-tuned for Arabic speech recognition and synthesis across dialects, supporting automated scoring. Bias mitigation includes auditing datasets for gender and dialect imbalances, using fairness-aware algorithms [8].

Integration Plan: The MVP demo integrates with HR platforms (e.g., Workable), job boards (e.g., Wuzzuf), Google Meet APIs for AI-led interviews, and Gmail's SMTP for email notifications. The full vision adds integrations with SAP SuccessFactors, Bayt.com,

and enhanced Google Meet APIs for video recording and automated scoring. AWS Media Insights will handle storage and analysis of video and audio recordings. Partnerships with AWS, Google Cloud, and regional job platforms ensure scalability, with initial focus on Egypt (MVP by Month 9) and expansion to Saudi Arabia and Jordan (Month 18).

The roadmap ensures the MVP validates core functionality, including voice-based Google Meet interviews, by Month 9, with the full vision—multi-dialect support, automated scoring, voice, and video—deployed by Month 18, positioning *SurgeAI* to meet the diverse needs of the MENA recruitment market.

4 Business Model & Strategy

4.1 Business Model Canvas

The Business Model Canvas (BMC) for *SurgeAI* outlines the strategic framework for delivering an AI-powered interviewer platform tailored for Arabic users in Egypt. Developed by Osterwalder [15], the BMC organizes nine building blocks to define how *SurgeAI* creates, delivers, and captures value. The platform serves corporate HR departments with automated candidate screenings, individual job seekers with mock interview practice, and universities with employability tools, all customized for the Egyptian dialect. The canvas, shown in Figure 2, provides a concise overview, with detailed explanations below.

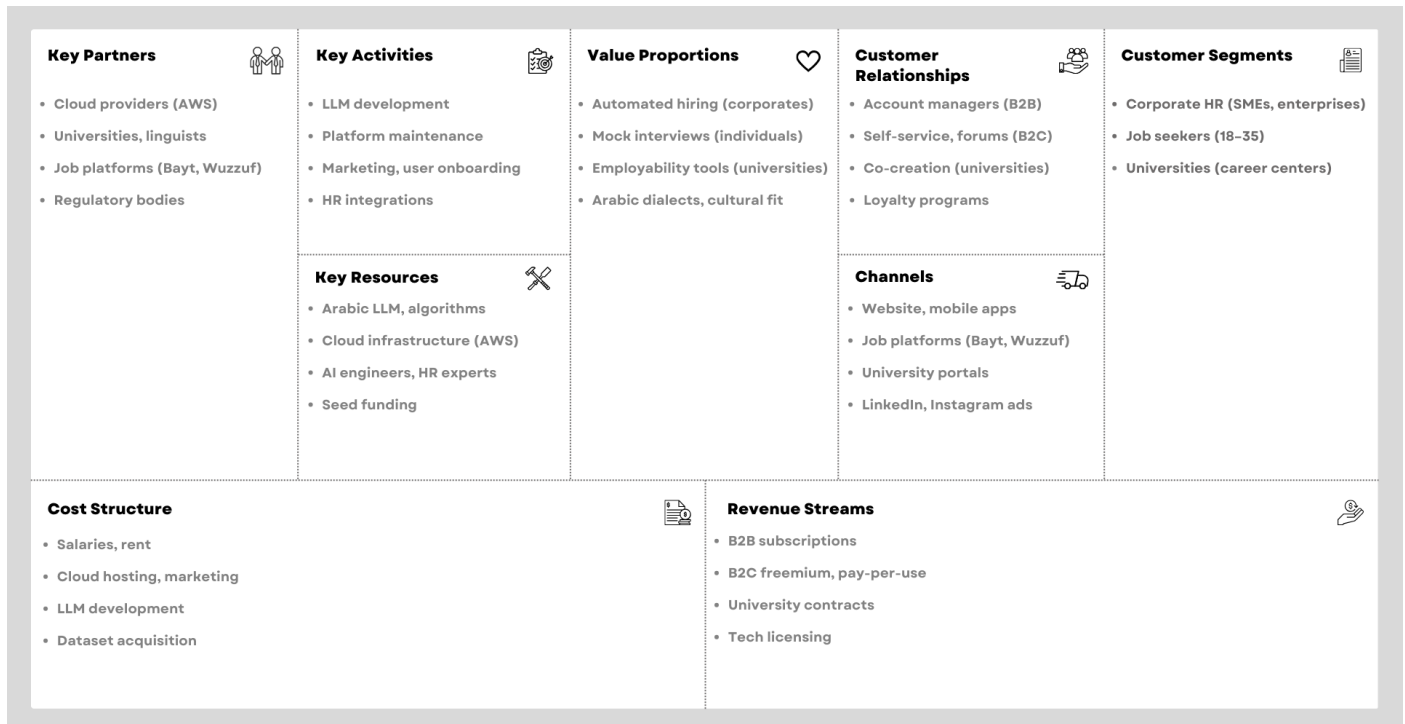


Figure 2: Business Model Canvas for SurgeAI

Customer Segments: *SurgeAI* targets three primary segments: corporate HR departments in SMEs in Egypt, focusing on sectors like technology, manufacturing, and finance; individual job seekers aged 18–35, including students and professionals in urban centers (e.g., Cairo, Alexandria); and universities with career centers (e.g., Cairo University, Ain Shams University) serving students aged 18–25. These segments reflect Egypt’s growing workforce and youth employment initiatives.

Value Propositions: For corporates, *SurgeAI* automates initial screenings, reducing hiring time and costs with customizable, culturally relevant questions and manual scoring. For individuals, it offers text-based mock interviews with personalized feedback, tailored to Egyptian workplace norms, enhancing job readiness. Universities benefit from affordable tools to prepare students for competitive job markets, aligning with Egypt’s employability goals. The unique proposition is its Egyptian dialect support and affordability, absent in global competitors like HireVue [11].

Channels: *SurgeAI* reaches customers through direct channels, including its website (www.surgeai.com, hypothetical), and indirect channels via partnerships with Egyptian job platforms (e.g., Wuzzuf, Forasna) and university career portals. Awareness is driven by digital marketing on Instagram and Facebook (for job seekers) and LinkedIn (for HR professionals), supplemented by presence at local tech events like Cairo ICT.

Customer Relationships: Corporates receive automated support for SMEs, with email assistance for setup. Individuals access self-service via the web platform, with community forums for peer learning. Universities engage in co-creation, tailoring programs to curricula, with periodic feedback. Referral programs (e.g., \$1 credit per invite) foster retention across segments.

Revenue Streams: *SurgeAI* generates revenue through B2B subscriptions for corporates (\$20–\$200/month based on interview volume), a B2C freemium model (free basic interviews, \$3/month premium, \$1 pay-per-use), and B2B2C university contracts (\$1,000–\$2,000/year). Additional revenue may come from licensing AI technology.

Key Resources: The platform relies on a fine-tuned Arabic LLM (e.g., AraBERT), cloud infrastructure (e.g., AWS), and a lean team of AI engineers, marketers, and Egyptian cultural experts. A hypothetical \$100,000 grant supports initial operations.

Key Activities: Core activities include fine-tuning the Arabic LLM, maintaining the web platform, integrating with HR systems, and executing low-cost marketing campaigns. Continuous user feedback analysis ensures platform relevance.

Key Partners: *SurgeAI* collaborates with cloud providers (e.g., AWS) for hosting, universities (e.g., Cairo University) for datasets, job platforms (Wuzzuf, Forasna) for distribution, and Egyptian regulatory bodies (e.g., Ministry of Communications) for compliance. Student ambassadors enhance outreach.

Cost Structure: Fixed costs include salaries (\$28,000/year for 5 staff), Cairo co-working space rent (\$2,400/year), and software licenses (\$2,000). Variable costs cover cloud hosting (\$5,000/year), marketing (\$7,000/year), dataset acquisition (\$3,000/year), and compliance (\$2,000/year). Initial LLM development (\$15,000) is a capital cost. Cost optimization leverages open-source tools and university partnerships.

4.2 Go-To-Market Strategy

SurgeAI's go-to-market (GTM) strategy aims for rapid penetration in Egypt, with plans to expand to Saudi Arabia, using low-cost marketing, sales, and distribution to reach corporate HR, job seekers, and universities.

Marketing Strategy:

- *Digital Campaigns:* Target Instagram and Facebook (job seekers aged 18–35) and LinkedIn (HR professionals) with ads highlighting affordability and Egyptian customization. Budget: \$5,000/year.

- *Content Marketing*: Share blogs and social media posts on AI recruitment, hosted on the *SurgeAI* website and Wuzzuf. Example: “Affordable AI Hiring in Egypt” post.
- *Events*: Leverage student ambassadors at universities and attend Cairo ICT to demo *SurgeAI*. Cost: \$2,000/year.
- *PR*: Feature in local outlets like Daily News Egypt to highlight first-mover advantage in Arabic AI interviews.

Sales Strategy:

- *B2B*: Pitch to SMEs with 30-day free trials to demonstrate ROI. Target: 10 corporate clients in Year 1, focusing on tech and manufacturing.
- *B2C*: Promote freemium model via website and social media, with referral programs (\$1 credit per invite). Target: 1,000 users in Year 1.
- *B2B2C*: Negotiate university contracts, starting with pilots at Cairo University. Target: 1–2 partnerships by Year 2.

Distribution Strategy:

- *Direct*: Deliver via web platform, ensuring 99.9% uptime with AWS.
- *Partner*: Integrate with Wuzzuf and Forasna for job seeker and recruiter access.
- *University*: Embed in career portals, training staff for adoption.

The GTM strategy leverages digital channels and partnerships to achieve 1,000 individual users and 10 corporate clients by Year 1.

4.3 Financial Projections

SurgeAI’s financial projections estimate startup costs, pricing, revenue, and breakeven, based on Egypt’s recruitment trends.

Startup Costs:

- *Development*: \$15,000 (LLM fine-tuning with AraBERT), \$5,000 (AWS credits).
- *Operations*: \$28,000 (Year 1 salaries, 5 staff), \$2,400 (Cairo co-working space).
- *Marketing*: \$7,000 (digital, events).
- *Datasets*: \$3,000 (Arabic dialects via universities).
- *Legal/Compliance*: \$2,000 (Egypt Data Protection Law).
- *Total*: \$72,400 (hypothetical grant).

Pricing Strategy:

- *B2B*: Subscriptions: \$20/month (10 interviews), \$80/month (50 interviews), \$200/month (unlimited). Licensing: \$2,000.

- *B2C*: Freemium: free (2 interviews/month), premium \$3/month (10 interviews). Pay-per-use: \$1.
- *B2B2C*: Contracts: \$1,000–\$2,000/year.

Revenue Projections:

- *Year 1*: 10 corporates (\$6,000), 1,000 users (10% premium, \$3,600), 1–2 universities (\$1,500). Total: \$11,100.
- *Year 2*: 30 corporates (\$36,000), 5,000 users (15% premium, \$27,000), 5 universities (\$7,500). Total: \$70,500.
- *Year 3*: 80 corporates (\$144,000), 15,000 users (20% premium, \$108,000), 10 universities (\$15,000). Total: \$267,000.

Breakeven Analysis:

- *Year 1*: \$72,400 costs, \$11,100 revenue, -\$61,300 (grant-funded).
- *Year 2*: \$85,000 costs, \$70,500 revenue, -\$14,500.
- *Year 3*: \$110,000 costs, \$267,000 revenue, +\$157,000 (breakeven).

4.4 Risk Analysis

Technical Risks:

- *Limited Arabic Datasets*: Partner with Egyptian universities, invest \$3,000 in data curation, audit models.
- *Downtime/Security*: Use AWS (99.9% uptime), encrypt data, test biannually (\$1,000/year).

Market Risks:

- *Competition*: Leverage Egyptian dialect support, partner with Wuzzuf exclusively.
- *Low Adoption*: Offer free trials, freemium model, and social media campaigns. Price below global competitors [11].

Operational Risks:

- *Talent Shortage*: Recruit locally, offer \$3,000–\$8,000/year salaries, partner with universities.
- *Regulatory Non-Compliance*: Align with Egypt’s Data Protection Law, audit compliance (\$1,000/year) [1].

5 Implementation Plan

5.1 Development Timeline

The development timeline for *SurgeAI*, an AI-powered interviewer platform tailored for Arabic users, is structured into three phases: ideation, Minimum Viable Product (MVP), and full-scale launch. The MVP, defined as a web-based product with text-based mock interviews and manual scoring for the Egyptian dialect [19], enables early user feedback with minimal resources. The timeline spans 18 months, targeting Egypt with plans to expand to MENA.

Phase 1: Ideation (Months 1–3):

- *Market Research*: Validate demand through surveys with HR professionals (target: 30 responses) and job seekers (target: 100 responses) in Egypt. Cost: \$2,000.
- *Technical Feasibility*: Assess Arabic LLMs (e.g., AraBERT, CAMELBERT) for fine-tuning. Select AWS for scalability. Cost: \$5,000 for prototyping.
- *Business Planning*: Finalize Business Model Canvas, secure \$100,000 hypothetical grant from Egyptian accelerators (e.g., TIEC). Cost: \$2,000 for pitch preparation.
- *Team Recruitment*: Hire core team (5 part-time members: CTO, 2 developers, marketer, HR consultant). Cost: \$7,000 (initial salaries).

Phase 2: MVP Development (Months 4–9):

- *Product Development*: Build MVP with text-based mock interviews (B2C) and basic screening with manual scoring (B2B), supporting Egyptian dialect. Fine-tune LLM for question generation. Cost: \$20,000 (development, datasets).
- *User Interface*: Develop web interface (React, Next.js) with intuitive UX for HR and job seekers. Cost: \$5,000.
- *Testing*: Conduct alpha testing with 20 job seekers and 3 SMEs in Cairo. Iterate based on feedback (e.g., dialect accuracy, UX). Cost: \$2,000.
- *Partnerships*: Initiate pilots with one university (e.g., Cairo University) and one job platform (e.g., Wuzzuf). Cost: \$2,000 (outreach).

Phase 3: Full-Scale Launch (Months 10–18):

- *Feature Expansion*: Add automated scoring, Gulf dialect support, and HR integrations (e.g., Workable). Cost: \$15,000.
- *Beta Testing*: Launch beta with 200 job seekers, 5 corporates, and 1–2 universities in Egypt. Collect KPIs (e.g., user retention). Cost: \$5,000.
- *Marketing*: Roll out digital campaigns (Facebook, LinkedIn) and leverage student ambassadors. Cost: \$5,000.

- *Scaling*: Expand to Saudi Arabia, targeting 1,000 users and 10 corporate clients. Secure \$200,000 follow-on funding. Cost: \$5,000 (market entry).

The timeline ensures a lean approach, with the MVP validating core features by Month 9 and the full launch achieving market traction by Month 18.

5.2 Team & Resource Allocation

SurgeAI's implementation requires a lean team and strategic resource allocation to deliver the platform efficiently within Egypt's ecosystem.

Team Roles and Responsibilities:

- *Chief Technology Officer (CTO)* (1, part-time): Oversees LLM development and technical strategy. Salary: \$8,000/year.
- *Software Developers* (2, part-time): Build web platform and integrations. Salary: \$6,000/year each.
- *Marketing Manager* (1, part-time): Lead digital campaigns and outreach. Salary: \$5,000/year.
- *HR Consultant* (1, part-time): Advise on Egyptian recruitment norms and question design. Salary: \$3,000/year.
- *Total Team*: 5 members, Year 1 cost: \$28,000.

Resource Allocation:

- *Technology*: AWS free tier + \$5,000 credits, open-source tools (e.g., PyTorch, \$2,000).
- *Datasets*: Egyptian dialect datasets from universities (e.g., Cairo University, \$3,000).
- *Office*: Co-working space in Cairo (rent: \$2,400/year).
- *Marketing*: \$5,000 for digital ads, \$2,000 for events (e.g., Cairo ICT).
- *Legal/Compliance*: \$2,000 for Egypt Data Protection Law compliance.
- *Total Resource Cost*: \$21,400 (Year 1, excluding salaries).

The team and resources prioritize rapid MVP development and cost efficiency through open-source tools and university partnerships.

5.3 Milestones & Metrics

SurgeAI's progress is tracked through key milestones and performance indicators (KPIs), ensuring alignment with business objectives.

Milestones:

- *Month 3 (Ideation)*: Secure \$100,000 grant, complete market research (30 HR, 100 job seeker responses), recruit core team (5 members).
- *Month 6 (MVP Development)*: Complete LLM fine-tuning for Egyptian dialect, develop MVP (text-based interviews, manual scoring).
- *Month 9 (MVP Launch)*: Launch MVP to 20 job seekers and 3 SMEs, secure pilot with 1 university and 1 job platform.
- *Month 12 (Beta Testing)*: Achieve 200 beta users, 5 corporate clients, and 1–2 university partners. Collect feedback for enhancements.
- *Month 18 (Full Launch)*: Reach 1,000 users, 10 corporate clients, expand to Saudi Arabia, secure \$200,000 funding.

Key Performance Indicators (KPIs):

- *User Acquisition*: 200 beta users by Month 12, 1,000 by Month 18 (B2C); 5 corporate clients by Month 12, 10 by Month 18 (B2B).
- *Retention Rate*: 60% user retention (B2C) and 70% client renewal (B2B) by Month 18.
- *Revenue*: \$11,100 by Year 1, \$70,500 by Year 2, tracking toward breakeven.
- *Technical Performance*: LLM accuracy >85% for dialect-specific responses, platform uptime 99.9%.
- *Customer Satisfaction*: Net Promoter Score (NPS) >40 by Month 12.

These milestones and KPIs provide a clear roadmap for *SurgeAI*, ensuring technical excellence and market traction by Month 18.

References

- [1] AGBI, “Generative AI gains momentum in the Gulf and MENA,” 2023. [Online]. Available: <https://www.agbi.com> [Accessed: May 5, 2025].
- [2] Akhtaboot, “Job Platform for MENA Recruitment Services,” [Online]. Available: <https://www.akhtaboot.com> [Accessed: May 5, 2025].
- [3] Avature, “AI-Driven Recruitment and Talent Management Solutions,” [Online]. Available: <https://www.avature.net> [Accessed: May 5, 2025].
- [4] Bayt.com, “MENA’s Leading Job Board and Recruitment Platform,” [Online]. Available: <https://www.bayt.com> [Accessed: May 5, 2025].
- [5] Business Market Insights, “Middle East & Africa Staffing and Recruitment Market to 2030,” [Online]. Available: <https://www.businessmarketinsights.com> [Accessed: May 5, 2025].
- [6] Cambridge Core, “Exploring AI governance in the Middle East and North Africa (MENA) region,” 2024. [Online]. Available: <https://www.cambridge.org> [Accessed: May 5, 2025].
- [7] CVViZ, “AI Recruitment Software for Candidate Engagement and ATS Integration,” [Online]. Available: <https://www.cvviz.com> [Accessed: May 5, 2025].
- [8] Fast Company Middle East, “The Middle East scores big in building Arabic AI models despite challenges,” 2024. [Online]. Available: <https://fastcompanyme.com> [Accessed: May 5, 2025].
- [9] Grand View Research, “Artificial Intelligence Market Size, Share & Trends Analysis Report,” 2024. [Online]. Available: <https://www.grandviewresearch.com> [Accessed: May 5, 2025].
- [10] Grand View Research, “UAE Artificial Intelligence Market Size | Industry Report, 2030,” [Online]. Available: <https://www.grandviewresearch.com> [Accessed: May 5, 2025].
- [11] HireVue, “AI-Powered Video Interviewing and Assessments,” [Online]. Available: <https://www.hirevue.com> [Accessed: May 5, 2025].
- [12] InterviewBuddy, “Online Mock Interview Practice Platform,” [Online]. Available: <https://www.interviewbuddy.com> [Accessed: May 5, 2025].
- [13] Maximize Market Research, “AI Recruitment Market: Global Industry Analysis and Forecast (2024-2030),” 2020. [Online]. Available: <https://www.maximizemarketresearch.com> [Accessed: May 5, 2025].
- [14] Meta AI, “Arabic Language Support for AI Applications,” [Online]. Available: <https://www.meta.ai> [Accessed: May 5, 2025].

- [15] A. Osterwalder and Y. Pigneur, “Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers,” Wiley, 2010. [Online]. Available: <https://www.strategyzer.com> [Accessed: May 5, 2025].
- [16] Panorama RH, “Etat des lieux de la pratique du recrutement au Maroc,” 2024. [Online]. Available: <https://panoramarh.com> [Accessed: May 5, 2025].
- [17] PwC, “The potential impact of AI in the Middle East,” [Online]. Available: <https://www.pwc.com> [Accessed: May 5, 2025].
- [18] Qureos, “AI-Driven Recruitment Platform for the GCC,” [Online]. Available: <https://www.quireos.com> [Accessed: May 5, 2025].
- [19] Wikipedia, “Minimum Viable Product,” [Online]. Available: https://en.wikipedia.org/wiki/Minimum_viable_product [Accessed: May 5, 2025].
- [20] Wuzzuf, “Egypt’s Leading Job Platform for Recruitment and Job Matching,” [Online]. Available: <https://www.wuzzuf.net> [Accessed: May 5, 2025].